

复旦大学计算机科学与技术学院
2021 ~ 2022 学年第二学期期末考试答案
☐ A卷 ☒ B卷

课程名称: C++面向对象程序设计 课程代码: COMP130135
开课院系: 计算机科学与技术学院 考试形式: 闭 卷
姓 名: _____ 学 号: _____ 专 业: _____ a

一、选择题 (每题只选择一个答案; 如选多个, 不计分。每题3分, 总分24分)

1、 B 2、 B

3、 B 4、 D

二、程序阅读题 (每题5分, 共20分)

1、 (4 0)
2、 (12)

三、程序填空题 (每空4分, 共24分)

1、
① <u> *begin = 0, *end = 0 </u>
② <u> subvalue = arr[i] </u>
③ <u> maxvalue = subvalue </u>
2、
① <u> iMin = j </u>
② <u> list[i] < list[iMin] </u>
③ <u> list[j], list[iMin] </u>

四、编程题（共32分）

1. （12分）

参考代码如下

```
#include <vector>
#include <iostream>
#include <algorithm>
#include <iterator>
using namespace std;

template <class T>
void em_swap(T first, T last)
{
    //typename iterator_traits<T>::value_type temp = *first;
    auto temp = *first;
    *first = *last;
    *last = temp;
}

template <class _BT>
void em_reverse(_BT first, _BT last){
    _BT temp;
    while ((first!=last)&&(first!=--last))
    {
        em_swap(first, last);
        ++first;
    }
}
```

2、Integer.h (20分)

```
#ifndef ...
#define ...
#include <...>

class Integer{
private:
    int content_[500];

    int len;
public:
    Integer(){ len=1; memset(a,0,sizeof(a));}

    Integer(const int);

    Integer(const char*);

    Integer(const Integer & T);

    Integer & operator=(const Integer & n) ;

    friend Integer operator+(const Integer & T, const Integer & V) const; //两个大数之间的相乘运算
    friend Integer operator*(const Integer & T, const Integer & V) const; //两个大数之间的相乘运算
    friend istream& operator>>(istream & in, Integer & b) //重载输入运算符
    friend ostream& operator<<(ostream & out, Integer & b) //重载输出运算符
```

```
};
```

```
Integer::Integer(const int b){
```

```
    int c,d = b;
```

```
    len = 0;
```

```
    memset(a,0,sizeof(a));
```

```
    while(d > MAXN)
```

```
    {
```

```
        c = d - (d / (MAXN + 1)) * (MAXN + 1);
```

```
        d = d / (MAXN + 1);
```

```
        a[len++] = c;
```

```
    }
```

```
    a[len++] = d;
```

```
}
```

```
Integer::Integer(const char* s)
```

```
{
```

```
    int t,k,index,l,i;
```

```
    memset(a,0,sizeof(a));
```

```
    l=strlen(s);
```

```
    len=l/DLEN;
```

```
    if(l%DLEN)
```

```
        len++;
```

```
    index=0;
```

```
    for(i=l-1;i>=0;i-=DLEN)
```

```
    {
```

```
        t=0;
```

```
        k=i-DLEN+1;
```

```
        if(k<0)
```

```
            k=0;
```

```
        for(int j=k;j<=i;j++)
```

```
            t=t*10+s[j]-'0';
```

```
        a[index++]=t;
```

```
    }
```

```
}
```

```
Integer::Integer(const Integer & T) : len(T.len) //拷贝构造函数
```

```
{
```

```
    int i;
```

```
    memset(a,0,sizeof(a));
```

```
    for(i = 0 ; i < len ; i++)
```

```
        a[i] = T.a[i];
```

```
}
```

```
Integer & Integer::operator=(const Integer & n) //重载赋值运算符，大数之间进行赋值运算
```

```
{
```

```
    int i;
```

```
    len = n.len;
```

```
    memset(a,0,sizeof(a));
```

```
    for(i = 0 ; i < len ; i++)
```

```
        a[i] = n.a[i];
```

```
    return *this;
```

```
}
```

```
istream& operator>>(istream & in, Integer & b) //重载输入运算符
```

```
{
```

```
    char ch[MAXSIZE*4];
```

```
    int i = -1;
```

```
    in>>ch;
```

```
    int l=strlen(ch);
```

```
    int count=0,sum=0;
```

```
    for(i=l-1;i>=0;)
```

```
    {
```

```
        sum = 0;
```

```

        int t=1;
        for(int j=0;j<4&& i>=0;j++,i--,t*=10)
        {
            sum+=(ch[i]-'0')*t;
        }
        b.a[count]=sum;
        count++;
    }
    b.len =count++;
    return in;
}

ostream& operator<<(ostream& out, Integer& b) //重载输出运算符
{
    int i;
    cout << b.a[b.len - 1];
    for(i = b.len - 2 ; i >= 0 ; i--)
    {
        cout.width(DLEN);
        cout.fill('0');
        cout << b.a[i];
    }
    return out;
}

Integer Integer::operator+(const Integer & T, const Integer& V) const //两个大数之间的相加运算
{
    Integer t(V);
    int i, big; //位数
    big = T.len > len ? T.len : len;
    for(i = 0 ; i < big ; i++)
    {
        t.a[i] +=T.a[i];
        if(t.a[i] > MAXN)
        {
            t.a[i + 1]++;
            t.a[i] -=MAXN+1;
        }
    }
    if(t.a[big] != 0)
        t.len = big + 1;
    else
        t.len = big;
    return t;
}

Integer Integer::operator*(const Integer & T, const Integer & V) const //两个大数之间的相乘运算
{
    Integer ret;
    int i, j, up;
    int temp, temp1;
    for(i = 0 ; i < len ; i++)
    {
        up = 0;
        for(j = 0 ; j < T.len ; j++)
        {
            temp = V.a[j] * T.a[i] + ret.a[i + j] + up;
            if(temp > MAXN)
            {
                temp1 = temp - temp / (MAXN + 1) * (MAXN + 1);
                up = temp / (MAXN + 1);
                ret.a[i + j] = temp1;
            }
            else
            {
                up = 0;
                ret.a[i + j] = temp;
            }
        }
    }
}

```

```

        if(up != 0)
            ret.a[i + j] = up;
    }
    ret.len = i + j;
    while(ret.a[ret.len - 1] == 0 && ret.len > 1)
        ret.len--;
    return ret;
}
#endif

```

考核要点：

- 1)书写完整类的能力；
- 2)使用基本数据类型的能力：例如vector,list等
- 3) 头文件相关：#ifdef ... #endif
- 4)包含相关的头文件： #include <vector> ...
- 5)两个操作符重载： + , *(应该定义为友元函数)
- 6)输入输出重载： <<, >>

2、完整代码：

```

#include <iostream>
#include <vector>
#include <string>
#include <fstream>

class Student{
public:
    std::string name;

    std::vector<int> scores;

    int total;

    float avg;

    Student(std::vector<std::string> words){
        std::vector<std::string>::iterator it = words.begin();
        name = *it;
        ++it;
        total = 0;

        while(it!=words.end()){
            int sc = atoi( (*it).c_str() );

```

```

        scores.push_back(sc);

        ++it;

        total += sc;

    }

    avg = total /scores.size();

}

```

```

friend std::ostream& operator<<(std::ostream& os, const Student& astu);

```

```

};

```

```

std::ostream& operator<<(std::ostream& os, const Student& astu)
{
    std::string SEP="\t";

    os<< astu.name << SEP << astu.avg << SEP;

    for(int i=0;i<astu.scores.size();i++)

        os<<astu.scores[i] << SEP;

    return os;

}

```

```

class Students{

    std::vector<Student> items;

public:

    Students(){

        std::string line;

        std::ifstream fin("z:\scores.txt");

        while(getline(fin,line)){

            std::vector<std::string> words = Split(line,"\t");

            items.push_back(Student(words));

        }

        fin.close();

    }

}

```

```

void Sort(){

    if (items.size()<2) return;

```

```

        for(size_t i=0;i<items.size();++i){

            for(size_t j=i+1;j<items.size();++j)

            {

                if(items[i].total<items[j].total){

                    Student temp = items[i];

                    items[i] = items[j];

                    items[j] = temp;

                }

            }

        };

};

std::vector<std::string> Split(const std::string& s, const std::string& c){

    std::vector<std::string> v;

    std::string::size_type pos1, pos2;

    pos2 = s.find(c);

    pos1 = 0;

    while(std::string::npos != pos2){

        v.push_back(s.substr(pos1, pos2-pos1));

        pos1 = pos2 + c.size();

        pos2 = s.find(c, pos1);

    }

    if(pos1 != s.length())

        v.push_back(s.substr(pos1));

    return v;

}

friend std::ostream& operator<<(std::ostream& os, const Students& list);

};

std::ostream& operator<<(std::ostream& os, const Students& list)

{

    for(size_t i=0;i<list.items.size();++i)

```

```
        os<<list.items[i] << "\n";  
    }  
    return os;  
}
```

考核要点：

- 1)书写多个完整类的能力；
- 2)文件操作；
- 3)输出重载；
- 4)基本算法：排序；
- 5)基本输入输出组织能力。